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From Published Evidence to Analyzable Biomedical Cohorts with Scientific Agents

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Abstract

A biomedical study can report how often two biomarkers are low without releasing the paired measurements that a researcher needs for a new analysis. We ask which scientific uses a virtual cohort can support when the available evidence consists of published statistics. Given a target population and the researcher's questions, a scientific agent builds a model of how the requested measurements vary together and records the assumptions it adds. Sampling this model produces a table on which several analyses can be run. For iron status, two agents received the same evidence. One model came closer to the US survey estimate of how often hemoglobin and ferritin were both low. The other came closer on their continuous rank correlation. Both underestimated the target threshold odds ratio. Across two glycemia tasks, adding published relationship evidence reduced mean absolute discrepancy for five of six regression coefficients. In an earlier evaluation with reference results withheld until generation ended, neither case met its prespecified overall criterion. The construction record lets a researcher trace these differences to scientific modeling choices and identify which uses need additional paired measurements.

1 Introduction

A researcher wants to study iron status among US women aged 20–44. A published study provides hemoglobin bands, geometric mean ferritin concentrations within those bands, and counts of women jointly classified by the two biomarker thresholds [33]. These counts give the proportion with both biomarkers low and the strength of their association in the source sample. What if the researcher also wants to know how the two concentrations vary together across their full ranges? Two women in the same hemoglobin band can have different ferritin values. The published counts leave their ordering open. A change of study population raises another question about how much of the source relationship will carry over.

With the paired measurements in hand, the researcher could answer all three questions on the same records. Our aim is to make that workflow possible when the individual measurements are unavailable. The product is a virtual cohort accompanied by a description of how its records were constructed. The common structure matters. A decision to weaken the pairing between the two biomarkers changes the results of several analyses at once. Writing that decision down gives a researcher a way to examine its consequences.

We give a scientific agent the population, intended analyses, variable definitions and admissible evidence. The agent determines which findings can enter a numerical model, chooses how to connect the measurements where the evidence leaves room, and writes a sampling program. Each row generated from the program contains a set of measurements for one synthetic participant. Different requests can call for different models. The iron study uses a published cross-tabulation and concentrations by band. A glycemia study supplies BMI and glucose slopes from separate regressions adjusted for age and sex. In both settings, researchers can examine which parts of an analysis follow from the literature and which depend on choices made during construction.

The experiments follow the researcher's questions. We first use the same continuous iron records to estimate co-occurrence, a threshold association and continuous correlation. We then ask whether adding external evidence about variable relationships changes the accuracy of adjusted regressions in two glycemia cases. Finally, we examine which findings extend to lipid and glucose requests, including earlier cases whose reference

results were withheld until the models and tables were frozen.

2 Related work and problem definition

2.1 From available information to analyzable records

Filshtein et al. share fitted study models so a recipient can simulate complete cohorts for epidemiological analyses [1]. SynthIPD uses published survival curves and trial summaries to reconstruct time-to-event records with covariates [2]. SynDI combines internal individual data with external regression information across populations to fit a target regression [29]. Multilevel network meta-regression likewise relates aggregate evidence to individual-level models under population assumptions [3]. These approaches differ in what the recipient knows. Filshtein et al. supply the comparison for our HRS experiment because the published exports and the scientific analysis can be matched. In our external-evidence experiments, the agent must also choose the joint dependence structure needed to perform the requested analyses.

G-Sim combines LLM-authored simulators with numerical calibration and structural refinement [4]. LLMSynthor uses aggregate discrepancies to guide proposals for new records and evaluates mobility, e-commerce and household population synthesis [5]. SynthCraft studies healthcare synthetic-data workflows [6], and SPADA samples from a dependency graph fitted with observed tabular data [15].

KMGen combines agentic extraction of published Kaplan–Meier curves with trial-record information and a mechanistic sampler to generate patient records with survival and adverse events [36]. Jones et al. generate household populations from aggregate demographics without individual input data, use statistical feedback and examine household relationships not supplied as constraints [37,38]. These studies already connect language models, published or aggregate evidence and synthetic records. Our experiments examine how an agent authors joint biomarker models for different research requests, handling source populations, measurement scales and adjustment sets, and which scientific analyses their records support. They do not establish quality or scaling superiority over these systems.

2.2 System architecture

Figure 1 centers on the decision that changes the data. A scientific agent proposes a joint model, writes a program to sample it, and records which parts follow from published evidence and which it supplied. A numerical worker executes a trial of that program. The coordinator returns the checks allowed by the study. The agent can keep the model or revise it within the study budget. The output is a versioned generator and a cohort that other researchers can analyze.

The coordinator supplies the task, handles access to papers and tools, binds program versions to their results, and fixes the seeds for delivery. It can request scientific or implementation review. The agent chooses how to use an admitted finding, how to represent missing relationships, and how to sample the requested variables. When the protocol allows revision, the agent also decides what a diagnostic warrants. The generator runs in an isolated process with network access disabled. The research packet and access rules define what information the author may use.

Written instructions and execution controls serve different purposes. The project's authoring instructions ask an author to connect each numerical input to its source and to record unanswered questions. In Q5 the coordinator supplied a fixed set of interfaces describing the required artifacts and allowed feedback. In the iron study a separate adapter called the submitted program on the worker. These interfaces document what an author was asked to produce. The returned program, trial receipts and final decision document what was actually done.

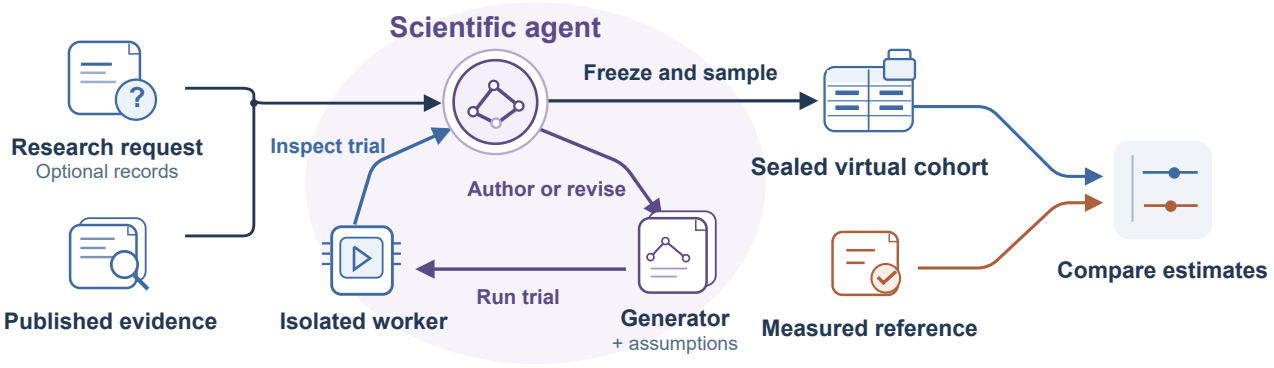


Figure 1. The scientific agent's action and feedback loop. The agent authors a generator from the research request and admissible evidence. An isolated worker runs the program and returns trial diagnostics for review. Depending on the protocol, the agent can retrieve literature during construction or revise its model after a trial. The chosen program generates the final cohort. In sealed evaluations, measured target results enter only on the comparison side. Supplement S12 identifies the implementation paths.

3 How a scientific agent constructs an analyzable cohort

3.1 Start with a population and a question

A request names the population, variables and scientific analyses, along with the observations and external studies that may be used. Knowing the intended analysis matters. Published counts of women with low hemoglobin and low ferritin already determine their co-occurrence and crude odds ratio in the source sample. A researcher who also wants to calculate a continuous rank correlation needs pairs of values, including within each threshold group. A researcher working in another population must consider how the pairing might change there.

For whole-cohort generation the agent receives a population description and generates all requested individual measurements. In our iron task, age and pregnancy status specify who the table represents. Only hemoglobin and raw ferritin are output columns. An enrichment request can instead supply observed covariates and ask for an additional variable conditional on those values. Partial completion can preserve observed cells and generate just the missing entries. The experiments in the main paper evaluate the whole-cohort setting.

The researcher determines the information boundary before construction. For the iron study, the two authors received identical external evidence, target eligibility and three intended analyses. They had no target biomarker values. In the summary-assisted Q2 arm, the protocol deliberately supplied target summaries. Comparing against a supplied statistic measures reconstruction. Comparing with previously withheld measurements asks whether the model also supports the analysis in that target. We report these conditions separately.

3.2 Decide which findings can enter the model

The agent can work with a reviewed packet or search for additional evidence when the protocol permits. In Q4 the authors started without bibliographic seeds and selected queries and original articles through Europe PMC/PMC. Each had up to eight query credits and four original papers. Four authors used twelve distinct originals in total. One author used six query credits and obtained four original articles. For example, the source review distinguished Bao et al.'s fasting glucose correlation with HbA1c (0.619) from the two-hour value (0.622) [7]. It also distinguished group medians and interquartile ranges from the standard deviation of individual measurements. These distinctions determine which numbers can enter a fasting-glucose generator.

The source review records the studies and statistics checked. In Q5, reviewed articles were supplied and further retrieval was optional. In the iron task, the coordinator had already qualified aspects of the evidence packet, including the contextual role of the Cuban study.

Table 1. Who decided how to construct the iron-status cohort? The two authors received the same request and evidence packet.

Contribution	Iron-status example
Investigator	Defines US eligibility, two requested measurements and three analyses. Identifies the corrected-ferritin result as contextual.
Published study	Supplies joint threshold counts and ferritin geometric means for three hemoglobin bands [33].
Agent A	Chooses a continuous model for each band. In the target law, it reduces the relative weight of low ferritin and independently redraws hemoglobin in one fifth of rows.
Agent B	Chooses a dependence model for ranks within bands. Carries the source joint distribution into the target context.
Numerical tools	Sample paired measurements from each frozen model. Apply all three analyses to each delivered table.

The agent records a source, study population, measurement definition, units and adjustment set for each quantitative use. A reported quantity can fix a model feature, help derive a parameter, or provide a check. A relevant paper with an incompatible measurement can be retained as context. A quantity the sources leave unspecified calls for a modeling assumption. Table 2 shows these decisions for the two continuous iron columns.

Table 2. What a paper supplies and what the agent must choose. The iron authors had access to the same source packet and task.

Available information	Consequence for construction
Zhu et al. report 53, 205 and 1,413 women in hemoglobin bands below 10, from 10 to below 12, and at least 12 g/dL, respectively [33].	The band proportions constrain how often a source draw belongs to each band. The author still chooses a continuous concentration distribution inside each band.
Ferritin geometric means in the bands are 1.3, 5.9 and 22.3 µg/L. The published proportions below 15 µg/L are 94.3%, 60.0% and 23.4%.	The source model can match mean log ferritin and depletion frequency by band. Several different ways of pairing continuous values can meet both constraints.
Joint threshold counts and band summaries come from the same participants.	The four counts give a crude odds ratio of 6.65. This follows algebraically from the counts.
The Cuban association uses ferritin corrected for inflammation [34].	The packet identifies its different measurement definition. Both authors retain it as context when modeling raw ferritin.
Neither source supplies the continuous rank correlation in the target population.	The author chooses within-band dependence and how much of the source model to carry into the target.

Unit conversions and population differences are recorded separately. The source hemoglobin threshold of 120 g/L becomes 12 g/dL by arithmetic. Carrying a ferritin distribution from the selected Chinese plasma study into a US survey of nonpregnant women requires assumptions about selection and measurement. Converting hemoglobin units resolves only the unit difference.

3.3 Write a joint sampling model and generate one row

For model B, generating a row begins with a draw from the three source hemoglobin bands in their reported proportions. The program draws a continuous hemoglobin value within the selected band and determines its rank there. It then draws a ferritin rank conditional on that hemoglobin rank. The chosen dependence rule is a copula. The program converts the ferritin rank into a positive concentration using a distribution on the log scale. Its probability of falling below 15 µg/L and its mean log concentration reflect the statistics for the selected band. The paired numbers are placed in the same row (Figure 2).

Two generators can assign the same proportion of low ferritin to a band and match its geometric mean, yet disagree on whether a woman with higher hemoglobin in that band also has higher ferritin. B supplied a within-band rank rule and applied the source distribution unchanged in the target context. A used a different continuous construction and made two changes for the broader target population. It gave low ferritin a relative weight of two thirds, then redrew hemoglobin independently for one fifth of target rows from the reweighted target hemoglobin distribution. Those fractions were A's assumptions. Section 5.1 shows their consequences for the threshold odds ratio and the joint proportion.

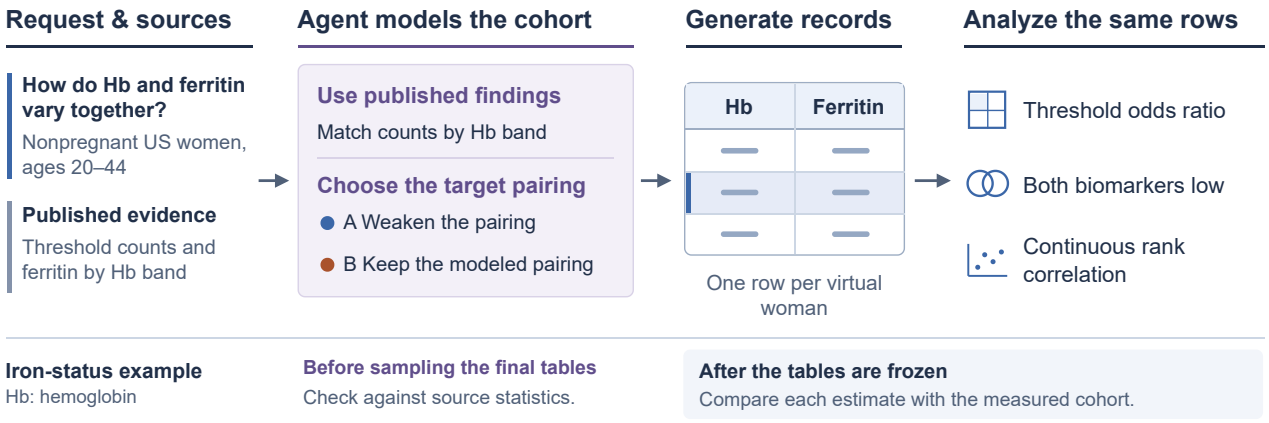


Figure 2. The iron-status request through the system. Study findings constrain the model that the agent builds. The agent also chooses how to pair measurements and how to apply the source evidence in the requested population. Each virtual row pairs hemoglobin with ferritin, so the three analyses use the same records. In the sealed iron evaluation the measured target biomarkers became available only after generation.

The common method is to build and record a sampling rule suited to the requested analysis. Its statistical form changes with the evidence. For CHNS glycemia, one constructor connected two separately adjusted source regressions by supplying a residual dependence assumption (Supplement S11). For HRS, fitted model exports supplied a starting point for generating repeated visits per person. Each construction delivers its own specification, numerical uses of evidence and executable program. A second author can produce a different joint law from the same packet.

3.4 Use trial feedback to make a scientific decision

The coordinator binds the candidate program to a particular version. A numerical worker samples it and reports the diagnostics the protocol allows. In a source replay the worker draws records under the source population rules and calculates the quantity reported in the source study, using its definition and units. A summary of a trial in the target context answers a different question about what that generator produced. A numerical comparison with the paper requires a compatible definition, weighting scheme and source population.

The iron task allowed one source-only review per author. For A, a trial of 100,000 source rows gave a crude odds ratio of 6.58, compared with 6.65 from the published counts. The simulated band-specific geometric

means also tracked the reported values. A separate inspection of the sampling rule found that the original upper ferritin tail had infinite variance. A finite trial can look plausible even when very rare draws make the model variance diverge. For concentrations above the cutoff, A used a rescaled square root of the original random draw on the log scale to slow the growth of the upper tail. The replacement preserves the mean log excess, threshold membership and rank order for fixed latent draws, and gives finite variance. This change repaired the mathematical behavior of the generator. The three registered analyses depend on thresholds or ranks, so the repair preserves their results for the same draws.

B's source trial produced an odds ratio of 6.73, with the three band means close to their source values. B retained the initial program. When a source statistic cannot be reproduced under its stated definition, the system records an unavailable check. In Q5 all 132 proposed source comparisons were unavailable because weighting, statistical definitions or details required for implementation were missing or unsupported. For example, the weight convention for a source summary of mean BMI in men was unspecified. The mean in an equally weighted synthetic sample could not stand in for that source estimator. The six Q5 authors could inspect their generated summaries and all retained their programs.

3.5 Freeze the tables and evaluate a scientific use

Once a final model has been selected, the coordinator records the program identity and draws all prespecified deliveries. In the iron study each model supplied 5,000 rows at each of three seeds. The program and table hashes were sealed before the measured target biomarkers were accessed. The evaluator ran the three prespecified analyses on the same two columns in every synthetic delivery. It computed the survey quantities using the eligibility, phlebotomy weight, strata and primary sampling units of the measured data. For Q5, the evaluator instead read previously computed reference estimates after the batch was closed. Supplement S12 documents the study-specific implementation paths.

An analysis can be rerun from a delivered table without asking the agent to refit the model to that result. For a threshold analysis the evaluator classifies each generated and measured pair using the same cutoffs. For continuous dependence it uses the original paired concentrations and their ranks. Agreement on threshold counts leaves room for differences in continuous ranks. Agreement for a named analysis is therefore evidence about that use of the table, not a certificate for its entire joint distribution or every later analysis. We also report marginal and residual variation where available.

The construction, the generated table and the measured participant are different units. Three draws from one frozen model show finite-table sampling variation conditional on that model. Independently authored models show variation across construction decisions; their spread is not a calibrated interval for model uncertainty. Measured participants contribute to the reference estimate under the cohort or survey design, including clustering and repeated observations where applicable. Generating more rows does not add measured participants or reduce uncertainty in the source evidence. These sources of variation answer different questions and are shown separately in the results.

4 Evaluation design

Each experiment specifies what the constructor can see and which analyses the delivered table should support (Table 3). We summarize the iron case first because it asks three questions of one dataset. The glycemia evidence contrast then varies one input to construction. Across studies, a construction is a separately authored model. Repeated draws quantify sampling variation within that model. We report discrepancies in the units of the intended scientific analysis and preserve each study's original decision rule.

Table 3. What the constructor could know, and what the evaluation can establish. Q1–Q5 are the identifiers of the frozen experiments. The iron study and the earlier prospective tests have separate protocols. All listed tasks generate complete cohorts.

Study	Construction inputs	Constructions and final deliveries	Analysis
Iron status	External evidence; target eligibility; no target biomarkers	2 agent; 6 tables of 5,000	Threshold association, joint burden and continuous ranks
Q1: CHNS and ENSANUT	Basic evidence (E0); E0 plus external relations (E1)	20 agent + 4 prescribed; 72 tables of 3,000	Six adjusted glycemia coefficients
Q5: glycemia and lipids	Reviewed external papers, population description, optional retrieval	6 agent; 18 tables of 3,000	Six requested effects or correlations
Earlier prospective wave	External summaries and population description; exact target results withheld	5 agent + 1 prescribed per case; 36 tables of 5,000	HAALSI glucose and Turkana lipid analyses
Q3: longitudinal reconstruction	Published HRS fitted-model artifacts	1 agent + 1 reconstructed; 6 valid tables, 11,509 subjects × 4 visits each	Age–HbA1c interactions in memory trajectories
Q2 / Q4: supporting conditions	Q2: with or without 30 target summaries; Q4: open paper retrieval without bibliographic seeds	Q2: 6 agent laws / 18 distinct tables; Q4: 4 / 12	Reconstruction of supplied summaries; repeated open retrieval

CHNS (6,631 adults, ages 35–74) uses unweighted estimates for the selected sample. ENSANUT (6,931 adults, ages 25–79) assigns equal mass to eight sex-by-age cells. In Q1, each case and information condition has five fresh agent authors and one prescribed statistical law. We average absolute discrepancy over three draws within each law, then across laws within the arm. E0 and E1 authors are not paired. The prescribed law illustrates a statistical completion from the same information; it is not a competitive search over statistical methods.

The reference conditions differ. Q1–Q5 reused project-level references where applicable. For iron status we sealed the protocol and the six deliveries before acquiring the target biomarkers; NHANES was already known to the project. In the earlier prospective wave, a custodian withheld the exact HAALSI and Turkana results, although investigators knew related Turkana findings. We distinguish target sampling intervals, resampling stability bands and model-conditional intervals in the figures. None makes a sample estimate the true population value.

5 Results

5.1 Which analyses can use the same iron-status records?

Zhu et al. studied 1,671 parous women who used IUDs in northern China, two to four years postpartum [33]. The source counts imply a crude odds ratio of 6.65. The agents' source replays gave 6.58 and 6.73. Both programs used the hemoglobin bands and their ferritin summaries. The Cuban study used inflammation-corrected ferritin [34]. As specified in the shared packet, its odds ratio supplied context for the transfer assumptions (Table 1).

Figure 3 traces the implications of the frozen code. In A, giving low-ferritin observations a relative weight of two thirds preserves the threshold odds ratio, since the factor cancels in the cross-product of counts. It reduces the joint-low proportion from 10.35% to 7.67%. For one fifth of rows A then draws hemoglobin independently from the same target hemoglobin distribution. The resulting model implies a joint-low proportion of 6.73% and an odds ratio of 4.65. B retains the source joint-low proportion of 10.35% and odds ratio of 6.65. These values follow from the two authored models. The colored markers in panels c to e show estimates from the six previously sealed tables.

The survey reference used the 979 women with both measurements among 1,065 eligible positive-weight women [35]. Missing either measurement excluded 86 women (8.08% unweighted; 2.47% of eligible phlebotomy-weight mass). The reported estimates describe this weighted complete-case reference; extending them to all eligible women requires assumptions about the missing measurements (Supplement S9). Comparison on the log-odds-ratio scale was primary. All six delivered odds ratios (A 4.835–4.866, B 6.011–6.565) fell below the survey's lower confidence limit of 9.28 (estimate 17.79, 95% CI 9.28–34.11). A came closer on the joint-low proportion (6.66–6.88% versus 7.63%), while B came closer on the continuous Spearman correlation (0.391–0.430 versus 0.398). For B the joint-low proportion was 10.02–10.44%, and for A the correlation was 0.156–0.195. The sources reported no continuous rank coefficient. Their ferritin summaries by hemoglobin band constrain ranks indirectly, leaving within-band dependence to the author. The same two columns in each delivery supplied all three estimates.



Figure 3. How two transfer choices affect analyses on the same iron-status records. Panels a and b are a post hoc calculation from the source counts and frozen rules. Panels c to e show three sealed tables of 5,000 rows per law and the NHANES estimate with its design-based 95% interval. Low hemoglobin is below 12 g/dL and low raw ferritin below 15 µg/L. The two secondary analyses had no prespecified equivalence margin.

5.2 Which regressions improve with additional relationship evidence?

The iron case illustrates what a fixed source packet leaves open. The next comparison asks whether adding published associations to basic evidence helps constrain a model. Across CHNS and ENSANUT, the agent arms given these additional relationships (E1) had smaller mean absolute discrepancies than the basic-evidence

arms (E0) for five of six regression coefficients (Figure 4). For the CHNS glucose slope the error fell from 0.134019 to 0.024079 HbA1c percentage points per mmol/L. The CHNS BMI error increased from 0.020380 to 0.031557 percentage points per kg/m². All three ENSANUT discrepancies decreased. These are descriptive contrasts between the sampled construction arms; they do not isolate a feedback or revision effect. Some E1 endpoints were closer under the prescribed statistical construction, including CHNS glucose (0.012021 versus 0.024079 for the agent mean) and ENSANUT BMI adjusted for glucose (0.007940 versus 0.025649). Table S1 reports both statistical arms and both agent arms for all six analyses.

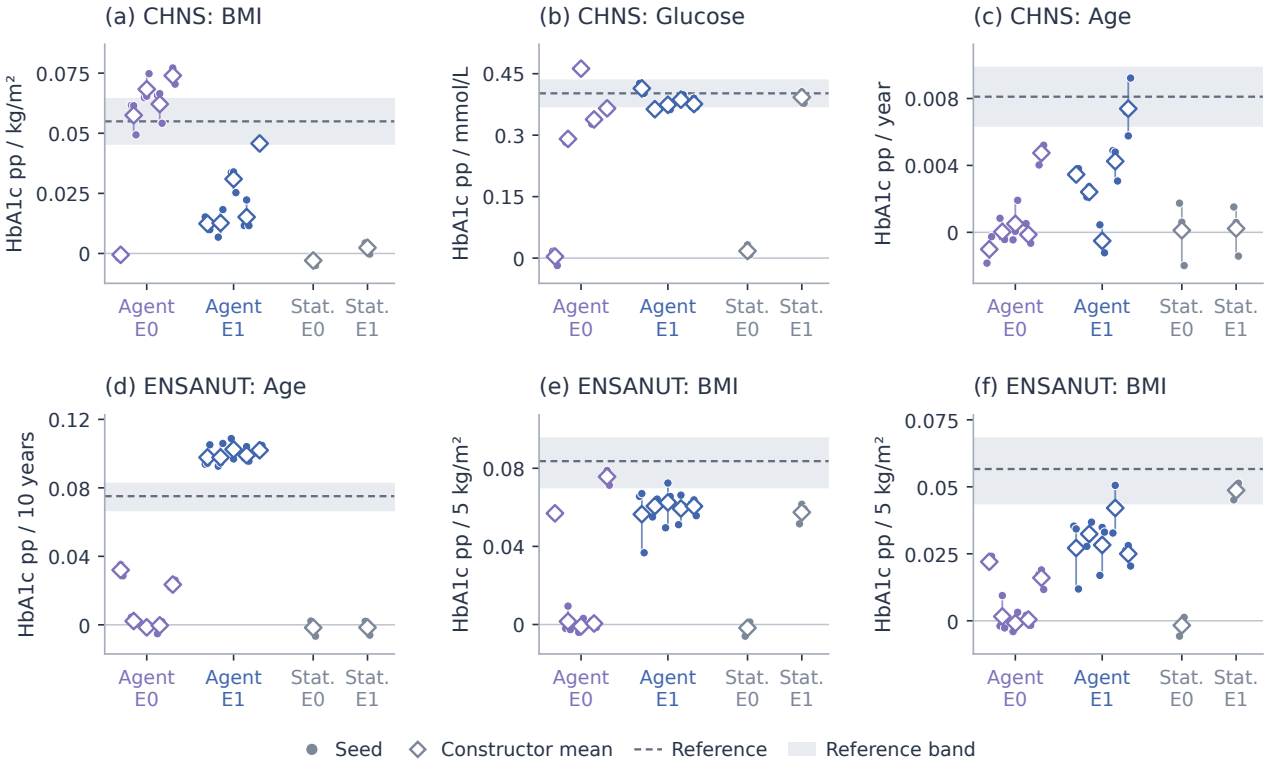


Figure 4. Relational evidence and scientific analysis agreement. CHNS panels show BMI adjusted for age/sex (a), glucose adjusted for age/sex (b), and age adjusted for sex/BMI/glucose (c). ENSANUT panels show age adjusted for sex/BMI/glucose (d), BMI adjusted for age/sex (e), and BMI additionally adjusted for glucose (f). All six analyses retain native units. E0 contains basic/marginal evidence; E1 adds relations. Each agent arm has five independently authored laws and each statistical arm one prescribed law, with all three final draws retained: 24 constructions, 72 tables and 216 coefficient estimates. Dots show seeds, open diamonds constructor means and thin segments seed ranges; dashed lines mark sample references. Reference bands are CHNS community-cluster CR1-t intervals or ENSANUT cluster-resampling stability bands. Author indices do not pair E0 with E1. Table S1 gives every endpoint's mean absolute discrepancy and E1–E0 contrast.

5.3 Which findings carry across requests and reference conditions?

For glycemia and lipids, three authors per request generated 18 tables in total. All 54 primary estimates agreed in point direction with the corresponding sample estimates across six distinct analyses (Figure 5). The tables supported age- and sex-adjusted BMI models and further adjustment for glucose. The three constructor-mean correlations of total and HDL cholesterol after adjustment for age, sex and BMI were 0.108–0.141, against 0.512 in the measured sample. The total-cholesterol slope's reference interval crossed zero. A post hoc check of the sealed lipid tables found HDL above total cholesterol in 69 of 27,000 synthetic rows (0.256%). Both concentrations were expressed in mmol/L. This check measures the biological consistency of individual pairs

separately from the prespecified association results.

A close coefficient did not imply that individual variation was reproduced. H03's mean glucose-adjusted BMI coefficient was 0.05098 versus 0.05664 HbA1c percentage points per 5 kg/m² in the reference. Its mean HbA1c SD was 0.3523 versus 0.5274 percentage points, and its full-model residual RMS was 0.3006 versus 0.4885 (Figure S3). These diagnostics qualify its regression-use result: the tables retained less marginal and residual variation. A study of individual heterogeneity or the tails would need additional validation.

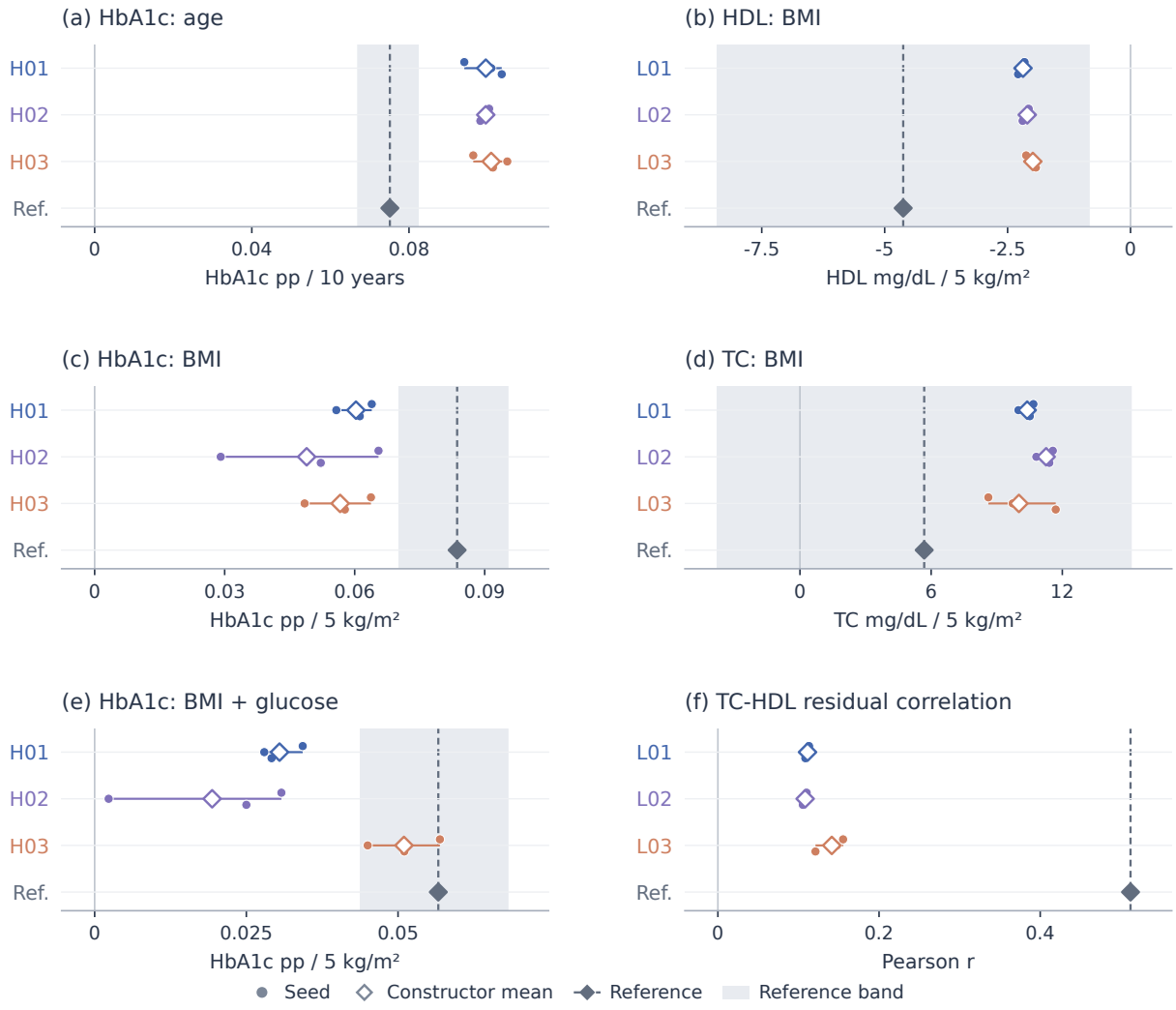


Figure 5. Analysis-specific behavior across glycemia and lipid requests. HbA1c panels in the left column show age adjusted for sex/BMI/glucose (a), BMI adjusted for age/sex (c), and BMI additionally adjusted for glucose (e). The right column shows age/sex-adjusted HDL and TC BMI slopes (b,d) and residual Pearson correlation after age/sex/BMI adjustment (f). All six laws and 54 estimates from 18 tables are retained. Circles show seeds, open diamonds means and thin segments seed ranges; dark diamonds and dashed lines mark sample references. HbA1c units are percentage points per 10 years or 5 kg/m²; lipid slopes are mg/dL per 5 kg/m². Bands are ENSANUT empirical cluster-resampling stability bands and lipid HC3 assumption-conditional sensitivity intervals. The TC band crosses zero; no residual-correlation interval is available. Seed ranges describe sampling under each frozen law. Table S5 supplies all numerical summaries and Figure S3 the dispersion results.

The earlier prospective wave tested a stricter question. In HAALSI adults aged 40 or older [30,31], all 15 agent draws recovered the positive directions of the adjusted BMI and age associations with capillary glucose;

only eight recovered the negative difference between female and male BMI slopes. The rule required four of five laws per case to meet all prespecified margins on every seed. Three passed in HAALSI and none in Turkana [32], so the planned expansion stopped. Even a zero-effect rule met the HAALSI margin for all three endpoints. The earlier method and its gate remain a separate prospective result, not a fresh evaluation of the later workflow on the same Turkana participants.

5.4 What changes when the scientific relationship is already supplied?

In the HRS longitudinal benchmark both methods received the published model exports [1]. Below the 6.5% HbA1c knot, the age–HbA1c interaction in memory trajectories averaged −0.04464 for agent tables and −0.04050 for the reconstructed published generator, against the supplied −0.04684. Above the knot the mean signs differed; five of six intervals included zero (Figure S6). This is a matched reconstruction check, not transfer from partial evidence. Similarly, supplying 30 target summaries in Q2 allowed joint reconstruction of the five highlighted regression and correlation analyses (Table S2). In Q4, open retrieval yielded twelve distinct original studies across four authors and twelve complete tables (Tables S3–S4); the design did not isolate the effect of retrieval. HRS and Q2 test reconstruction when target-model information is supplied; Q4 instead demonstrates a way to obtain external evidence without supplying more information about the target.

6 Discussion

For the researcher who asked about iron status, the same virtual cohort can yield a joint proportion, an odds ratio and a rank correlation. These results have different foundations. Published counts determine the threshold association in the source sample. The target odds ratio depends on how the agent changes the pairing across populations. For continuous correlation, concentrations within hemoglobin bands also matter. In this case, the law closer to the measured joint proportion differed from the law closer on continuous ranks. A researcher planning to study those ranks would gain information from paired concentrations within the bands in a comparable population. The lipid results point to an analogous need for joint measurements under the intended adjustment. These case comparisons show how to interpret results after evaluation. Assessing whether the method can flag a reliable use before seeing the reference would require a separate test.

A controlled diagnostic with exact, compatible summaries also found that ordinary constructors already reconciled separate projections into the same joint law; an additional consistency instruction gave no improvement (Supplement S13). This check supports keeping the workflow simple, while leaving noisy extraction and population transfer as separate empirical questions.

7 Limitations and Ethical Considerations

The evidence cannot identify every joint relationship. Sources differ in eligibility, age, geography, specimen handling and adjustment; reference samples have sampling uncertainty. The iron reference also conditions on both biomarkers being observed. Its small weighted missing fraction does not establish absence of selection bias, and no additional item-nonresponse sensitivity analysis was performed. A small number of agent constructions share a model service, and repeated draws do not represent independent clinical cohorts. The protocols differ across studies. No held-out trial shows that review improves outcomes or that the workflow saves a measured amount of human time. For research planning, the source-based and assumed parts should travel with the records. Values must be labeled synthetic and must not be presented as measured individual biomarkers, clinical diagnoses or substitutes for independent patient evidence. Original cohort access and reuse conditions continue to apply; the public artifacts contain code and aggregate results rather than participant-level tables.

8 Generative AI Usage

Language models participated in research planning and manuscript editing and were an object of the experiments. In the scientific workflow, agent authors interpreted evidence, selected or received studies according to protocol, proposed statistical models, and wrote or revised generator code. Deterministic tools sampled the records and computed the reported analyses; separate implementations checked the numerical results where specified. Study protocols and source records document these roles. The final author list, study-specific declarations, and applicable oversight and data-use statements must be confirmed before submission.

Supporting materials and reproducibility

The public research repository contains protocols, executable code and aggregate audits. The iron protocol seal and delivery seal precede target evaluation. Supplements retain all prespecified outcomes, unfavorable cases and implementation incidents. No new clinical cohort was generated for this revision. Supplement S13 reports a separate artificial mechanism diagnostic.

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Supplementary results

The supplement retains the complete numerical comparisons and supporting displays under their original study identities. Figure and table numbers follow their new placement; no construction, primary result or unavailable comparison is removed by the shorter main narrative. Q1–Q5 remain distinct experiments with their original information conditions.

S1. Complete external-evidence contrast

All 24 Q1 constructions produced their three scheduled final tables. The 72 final tables were distinct and supported every requested regression and diagnostic; seven exploratory tables brought physical executions to 79. Additional relational evidence reduced constructor-averaged mean absolute discrepancy for five of the six designated coefficients (Table S1). CHNS glucose discrepancy decreased from 0.134019 to 0.024079 HbA1c percentage points per mmol/L. All three ENSANUT contrasts improved. CHNS BMI discrepancy increased from 0.020380 to 0.031557.

Table S1. Complete main-study information and statistical comparisons. All numerical columns report mean absolute discrepancy in HbA1c percentage points per stated predictor increment, except the agent E1–E0 difference between those discrepancies. For each agent arm, absolute discrepancies are averaged over three draws within each of five constructions and then across constructions. Each statistical arm contains one prescribed construction and three draws. E1–E0 is calculated before rounding; E0 and E1 authors are not paired. The contrasts are descriptive. Rows retain their native units and are not pooled.

Case / focal analysis	Predictor increment	Agent E0	Agent E1	Agent E1 – E0	Statistical E0	Statistical E1
CHNS BMI, adjusted age/sex	1 kg/m²	0.020380	0.031557	+0.011178	0.057877	0.052555
CHNS glucose, adjusted age/sex	1 mmol/L	0.134019	0.024079	−0.109939	0.384750	0.012021
CHNS age, adjusted sex/BMI/glucose	1 year	0.007281	0.004854	−0.002426	0.007986	0.007878
ENSANUT age, adjusted sex/BMI/glucose	10 years	0.063944	0.024670	−0.039274	0.076797	0.076680
ENSANUT BMI, adjusted age/sex	5 kg/m²	0.056883	0.023732	−0.033151	0.085378	0.026202
ENSANUT BMI, adjusted age/sex/glucose	5 kg/m²	0.048797	0.025649	−0.023147	0.058345	0.007940

The effect displays explain what these changes mean. CHNS E1 glucose constructor means were 0.363363–0.414189 around the 0.402093 reference. Its BMI means were positive but lower, 0.012400–0.045757 versus 0.054946. Conditional-age means ranged from −0.000509 to 0.007393 versus 0.008107; one author retained a negative mean and two negative seeds. All ENSANUT E1 focal coefficients were positive, although its age means of 0.097769–0.102452 per decade exceeded 0.075132 and its BMI means remained below their references. Preserving a direction and recovering its magnitude are distinct aspects of usefulness.

Every E1 author used at least one relationship as a numerical constraint. The complete uptake records include constraints, unfitted analytic checks and contextual uses, including the case-specific projection calculation in Supplement S11. Statistical references also achieved useful results: under E1, the prescribed constructions

had lower seed-averaged discrepancies than every corresponding agent for CHNS glucose and ENSANUT conditional BMI. Other comparisons favored agent constructions. The evidence therefore supports scientific utility and the value of additional information, rather than a universal algorithmic ranking.

S2. Complete supplied-summary reconstruction

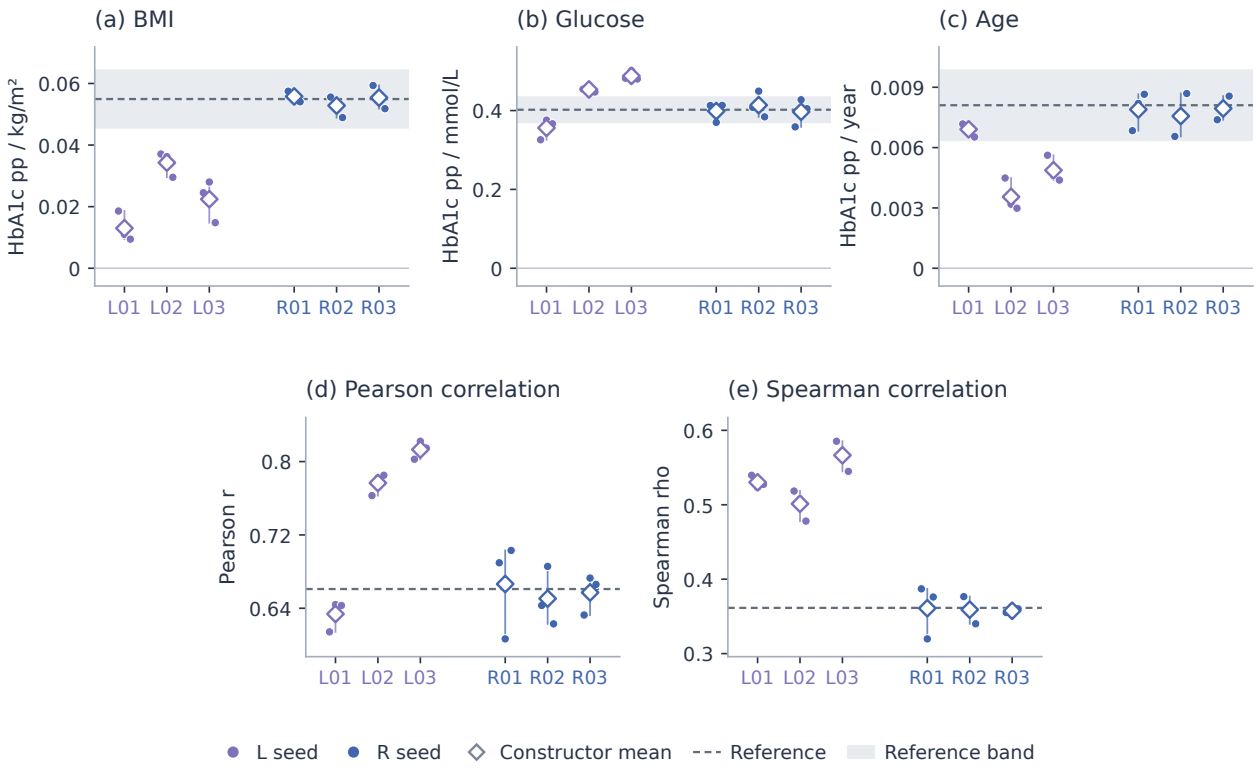


Figure S1. Supplied summaries become jointly analyzable data. Five panels show BMI, glucose and age regression coefficients plus glucose–HbA1c Pearson and Spearman correlations. L01–L03 use literature; R01–R03 additionally receive 30 target summaries and metadata. Points show all three final draws, diamonds constructor means and segments seed ranges. Dashed lines mark reference summaries available to R; regression bands are CHNS community-cluster CR1-t intervals, with no supplied correlation interval. All six constructors, 18 distinct delivery tables, 30 means and 90 seed statistics appear once. Initial and final deliveries were identical after retention; Table S2 gives the complete five-analysis numerical comparison.

The separate CHNS experiment gave three authors external literature (L) and three authors that literature plus 30 target summaries and their metadata (R). Each author delivered three tables and had one additional diagnostic-review opportunity. All six retained their models. There were 42 physical executions: six trials and 36 initial/final deliveries. Final bytes duplicated their matching initial outputs, leaving 24 distinct tables overall and 18 distinct delivery tables. The identical stages are displayed once.

R supported close reconstruction of all five highlighted scientific analyses (Table S2; Figure S1). These values were legitimate construction inputs; their agreement measures reconstruction rather than prediction of hidden coefficients. The L/R packet and authors differ from Q1, so those two information comparisons are kept separate.

Table S2. Joint-statistic reconstruction with available summaries. Ranges span three constructor means over three fixed seeds, not confidence intervals. Regression units are HbA1c percentage points per kg/m², mmol/L and year, respectively.

Analysis	Supplied reference	External literature L	Literature + summaries R
BMI coefficient, age/sex adjusted	0.054946	0.013001–0.034287	0.052821–0.055789
Glucose coefficient, age/sex adjusted	0.402093	0.356064–0.487815	0.396941–0.413715
Age coefficient, sex/BMI/glucose adjusted	0.008107	0.003552–0.006910	0.007569–0.007947
Glucose–HbA1c Pearson correlation	0.660968	0.633764–0.813170	0.650561–0.666438
Glucose–HbA1c Spearman correlation	0.361457	0.501339–0.566477	0.357571–0.360983

Agreement extended beyond central values. R constructor-mean BMI SDs were 3.38965–3.40243 versus 3.40632, glucose SDs were 1.48817–1.58968 versus 1.54455, and HbA1c SDs were 0.95208–0.96554 versus 0.95357. A separate three-table statistical calibration supplement also closely reconstructed the regression summaries, with mean coefficients 0.055726, 0.412568 and 0.007679. That purpose-built family was developed with coding-agent assistance and is not an agent-removal ablation. Both approaches demonstrated the usefulness of constructing a joint table from accessible aggregate information.

S3. Open retrieval across population requests

The unified workflow allocated two fresh scientific authors to each of the existing CHNS and ENSANUT requests. Each author received the intended analyses, population/measurement definitions, task interface and retrieval budget, but no target coefficients, desired signs, individual records or prior construction. CHNS- and ENSANUT-derived studies were excluded as generating evidence. Across 24 query credits, the authors selected sixteen paper uses representing twelve distinct originals. Independent review checked the original statistical definitions, numerical derivations, source applicability and required finite moments before execution. Sources could overlap across authors; four constructions do not imply four independent bodies of clinical evidence.

The agents chose materially different laws. C01 used an age-conditioned multivariate-lognormal mixture informed by Chinese population summaries [7,16] and contextual age/body-size findings [10,17]. C02 combined copula predictors with an affine HbA1c conditional mean using Pinggu summaries and Singapore coefficients [18,19], with other populations as applicability context [14,20]. E01 used bounded beta predictors and a conditional gamma response [10,17,21,22]. E02 combined lognormal BMI, conditional truncated-normal glucose and a positive conditional-mean lognormal response [10,21,23,24]. Their unreported dependencies, population transfers and residual choices remained declared assumptions. The shared controller did not prescribe these families or insert target coefficients.

Each author completed one 3,000-row trial, inspected generated-only diagnostics and source feedback, then selected its final model before three fixed draws. All four retained the initial law. The sixteen physical executions comprised four trials and twelve unique final tables, or 36,000 final virtual rows. Feedback contained 29 qualified approximate comparisons and 39 unavailable checks across the four trials, not source-population replays or 29 passed constraints. After all sessions closed, the evaluator reused the fixed sample references. A separate numerical implementation reproduced all 366 final coefficient/diagnostic values; no clinical data were downloaded or refitted.

All 36 primary seed estimates had the same positive directions as their corresponding references (Figure S2). These are repeated measurements of six analysis definitions across four laws, not 36 independent confirmations. Magnitude recovery varied by analysis. CHNS C01/C02 means were 0.03992/0.05338 for BMI against 0.05495, 0.37754/0.49241 for glucose against 0.40209, and 0.00311/0.00563 for age against 0.00811. Units are respectively HbA1c percentage points per kg/m², mmol/L and year. C02 was closer for BMI and age, while C01 was closer for glucose; no construction dominated all analyses.

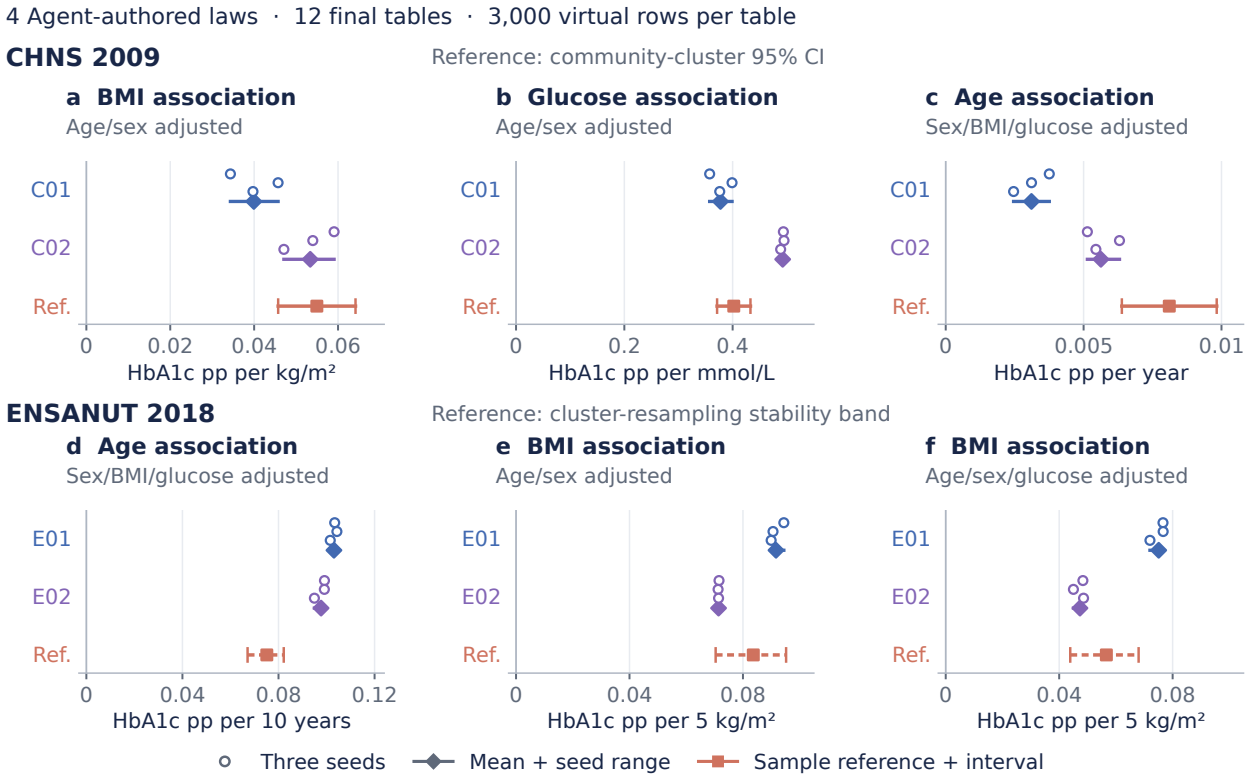


Figure S2. Repeated open-retrieval construction across two biomedical requests. All six prespecified primary analyses are shown for four agent-authored laws and twelve final 3,000-row virtual tables. Blue and purple identify the two constructions within each request. Open circles show seeds 8101–8103, ordered top to bottom within each author row; vertical spacing is solely for legibility. Diamonds and horizontal segments below the circles show the three-seed mean and minimum–maximum range. Coral squares in separate reference rows show previously exposed selected-sample estimates. Solid coral intervals are CHNS community-cluster CR1-t 95% confidence intervals; dashed coral intervals are ENSANUT empirical cluster-resampling stability bands from 399 attempts. These reference intervals have different meanings and neither is an acceptance or equivalence threshold. Axes retain native effect units and zero; pp denotes HbA1c percentage points. All generated coefficients are positive, while both CHNS age estimates are weaker and both ENSANUT age estimates stronger than their references. Seed ranges describe sampling under each fixed law, not confidence intervals or uncertainty over evidence, model choice or transport. The four constructions provide developmental workflow replication on two existing references, not independent clinical replication.

For ENSANUT, E01/E02 means were 0.10308/0.09772 for adjusted age against 0.07513 pp/decade, 0.09165/0.07140 for age/sex-adjusted BMI against 0.08364 pp per 5 kg/m², and 0.07509/0.04733 after additionally adjusting for glucose against 0.05664. Every draw retained the reduction of the BMI coefficient after glucose adjustment, an association pattern rather than causal mediation. Both authors built an age coefficient near 0.10 pp/decade and shared some sources; their agreement does not establish independent evidence convergence.

The complete diagnostics reveal remaining limits. CHNS C01 glucose–HbA1c Pearson correlation was approximately 0.666 versus 0.661, but Spearman correlation was 0.635 versus 0.361. ENSANUT residual HbA1c

RMS was approximately 0.295/0.324 versus 0.488, and total SD was 0.362/0.406 versus 0.527. Correct directions and a close Pearson coefficient therefore coexisted with rank and dispersion mismatches. The complete replication report, every diagnostic and Supplementary Table S3 retain the full results. A scalar-type interface defect in the original evaluator was corrected additively on unchanged frozen tables, with its failed output and independent audit preserved; no scientific model was adjusted after comparison.

Table S3. Complete Q4 open-retrieval primary summaries. Means and ranges summarize three final seeds under each fixed construction. Units are HbA1c percentage points per stated predictor increment. No row is omitted; ranges are not confidence intervals.

Request / analysis	Author	Increment	Reference	Generated mean	Seed min–max
CHNS BMI	C01	1 kg/m²	0.054946	0.039923	0.034342–0.045708
CHNS glucose	C01	1 mmol/L	0.402093	0.377538	0.357560–0.398892
CHNS adjusted age	C01	1 year	0.008107	0.003109	0.002461–0.003755
CHNS BMI	C02	1 kg/m²	0.054946	0.053375	0.047088–0.059055
CHNS glucose	C02	1 mmol/L	0.402093	0.492412	0.488649–0.494875
CHNS adjusted age	C02	1 year	0.008107	0.005628	0.005137–0.006301
ENSANUT adjusted age	E01	10 years	0.075132	0.103084	0.101546–0.104315
ENSANUT BMI	E01	5 kg/m²	0.083643	0.091651	0.089940–0.094413
ENSANUT glucose-adjusted BMI	E01	5 kg/m²	0.056638	0.075089	0.072005–0.076661
ENSANUT adjusted age	E02	10 years	0.075132	0.097724	0.094947–0.099135
ENSANUT BMI	E02	5 kg/m²	0.083643	0.071399	0.071232–0.071563
ENSANUT glucose-adjusted BMI	E02	5 kg/m²	0.056638	0.047325	0.045022–0.048592

CHNS BMI/glucose analyses adjust for age and sex; CHNS age adjusts for sex, BMI and glucose. ENSANUT age adjusts for sex, BMI and glucose; its BMI analyses adjust for age/sex, with or without glucose. All Q4 diagnostic values remain in the complete diagnostic table.

Table S4. Original-source inventory for Q4 open retrieval. Twelve unique originals supplied sixteen author-specific paper uses. Sharing a source is not independent clinical evidence; source populations are not the target populations.

Original	Reference	Constructing authors
Wu et al., 2017	[16]; PMC5591004	C01
Bao et al., 2010	[7]; PMC2871988	C01
Pani et al., 2008	[10]; PMC2551641	C01, E01, E02

Table S4 continued

Original	Reference	Constructing authors
Skogberg et al., 2019	[17]; PMC6458679	C01, E01
Zhang et al., 2016	[18]; PMC5078665	C02
Venkataraman et al., 2012	[19]; PMC3504343	C02
Liu et al., 2016	[20]; PMC5009131	C02
Wisgerhof et al., 2020	[14]; PMC7272077	C02
Huang et al., 2021	[21]; PMC8038122	E01, E02
Ghazanfari et al., 2010	[22]; PMC3075530	E01
Masuch et al., 2019	[23]; PMC6371438	E02
Marini et al., 2017	[24]; PMC5398507	E02

The reviewed evidence/specification files distinguish reported statistics, algebraic derivations and assumed relations. The public experiment record includes per-source eligibility, measurement, adjustment, overlap and reporting-quality notes; full third-party originals and participant-like rows remain local.

S4. Longitudinal reconstruction from published artifacts

The HRS recipient-stage example [1] tested a different data product. A fresh agent received published model exports, a variable specification and the intended mixed-model analysis. It authored the recursive conditional generating family with parameter provenance. A separate implementation reconstructed the published law. Each generated 11,509 subjects at 0, 2, 4 and 6 years, producing 46,036 rows per table. No HRS individual records were used, and the source analysis coefficients were construction inputs.

All three agent and three corrected-law tables supported the native R/lme4 model with 16 fixed effects, individual intercepts/slopes and piecewise HbA1c terms around 6.5%, with age centered at 60 and scaled per decade. The below-threshold HbA1c-by-age interaction averaged −0.044635 in agent data, with range −0.051059 to −0.038247, against the supplied −0.046837. The reconstructed law averaged −0.040500. Both preserved its negative direction in all three draws (Figure S6).

Figure S6 shows the estimates and their per-table Wald intervals. The longitudinal analysis record gives all 16 fixed effects and the comparator correction.

The weaker above-threshold interaction was less stable: agent estimates averaged 0.026485 against 0.012867, while corrected-law estimates averaged −0.011691. Overall agent HbA1c mean/SD was 5.8725/0.9676 versus the original published 5.9/1.0. The high-HbA1c subgroup had mean/SD 7.1515/0.5630, compared with original 7.7/1.4 and published synthetic 7.2/0.6. Thus useful source-model behavior coexisted with inadequate upper-group spread.

A CRLF-sensitive parser initially overwrote seven coefficient/SE pairs in our comparator preparation. A separately reviewed source-bound repair produced the three corrected comparator tables without changing the frozen agent outputs. The full record contains nine physical generations, six valid comparison tables and three invalid initial comparator attempts. This was our implementation defect, not a failure of the published method. The incident record and complete effects retain the correction and all outcomes.

S5. Cross-task construction, dispersion and source uses

A post hoc biological consistency check reports the HDL and total cholesterol ordering counts by frozen law and seed. The comparison uses generated pairs in their original common units.

The next study applied the common workflow to HbA1c analyses in the ENSANUT2018 analytical design and BMI-related TC/HDL analyses in the Turkana study context. Three fresh agent contexts constructed a law for each request, using reviewed external articles, population/measurement metadata and optional literature tools. No participant inputs, target coefficients, target means or prior generators were supplied. The six contexts used the same model service and shared some source evidence. This study therefore tests repeated construction across two requests, with the law chosen by each author, rather than independent clinical replication or a controlled comparison of workflow components.

The HbA1c analyses retained the existing age and BMI estimands and equal allocation to eight sex-by-age cells. The lipid analyses used unweighted age/sex-adjusted BMI coefficients for TC and HDL, plus the Pearson correlation of their residuals after age, sex and BMI adjustment. Lipid eligibility required adults with recognized recorded sex and finite, positive BMI and lipid values. Generated lipid concentrations were converted from mmol/L to mg/dL using the frozen factor 38.67. The reference represents a selected Turkana/Laikipia study sample; fasting, assay transfer and population composition remained qualified in the evidence record. No new participant records were read, downloaded or refitted for this study.

The agents made different scientific choices within the common interface. H01 combined bounded predictors, conditionally truncated glucose and a shifted gamma HbA1c innovation; H02 used bounded beta conditional laws. H03 solved cross-source moment equations under declared dependence and dispersion assumptions, then used conditional beta laws. Their source records included overlapping age, body-size and glycemia evidence [10,17,21,22], with additional check-only/contextual evidence for H01 [14]. The lipid authors combined Gaussian copulas and lognormal margins using geometric summaries from Greiner et al. [25], rural Ashanti rank correlations [26] and arithmetic moments from Agongo et al. [27,28]. Choices about arithmetic versus geometric location, transfer of coefficients of variation, sex strata and age dependence differed among L01–L03. These are applications of established probability families whose numerical use was selected and explained by the agent.

Independent source and implementation reviews checked the statistical meanings, specification/code consistency and finite moments before isolated execution. Each constructor completed one scheduled trial, inspected generated diagnostics and source feedback, and could revise once. All six retained their initial laws. Three fixed final seeds, 9101–9103, produced 18 distinct 3,000-row tables: 54,000 fictional rows from 24 successful physical attempts, including the six trials. No final execution failed and no seed was replaced. Reference comparison began after all six author sessions closed and reused previously exposed aggregate results.

All 54 primary point estimates had the same direction as their corresponding sample estimates, with quantitative agreement varying by analysis (Figure 5; Table S5). HbA1c age effects averaged 0.09954–0.10091 percentage points per decade versus 0.07513 in the reference; age/sex-adjusted BMI effects averaged 0.04892–0.06030 versus 0.08364 per 5 kg/m². With glucose additionally adjusted, H03 averaged 0.05098 versus 0.05664, whereas H01 and H02 averaged 0.03047 and 0.01936. Thus the shared age direction and attenuation after glucose adjustment did not imply recovery of each effect magnitude.

For lipids, mean HDL BMI coefficients ranged from −2.1865 to −1.9868 mg/dL per 5 kg/m² versus −4.6230 in the reference, while TC coefficients ranged from 10.0151 to 11.2609 versus 5.6831. The TC reference sensitivity interval was −3.7535 to 15.1197, so matching its positive point estimate does not establish replication of a definite positive population effect. Residual TC–HDL correlations averaged 0.10828–0.14134 versus 0.51207. The generators preserved the directions of these focal point estimates while substantially underrepresenting the lipid dependence remaining after the specified adjustment.

Dispersion changes the assessment of usefulness (Figure S3). H03's closer glucose-adjusted BMI coefficient accompanied lower HbA1c marginal SD and full-model residual RMS, 0.3523 and 0.3006 percentage points versus 0.5274 and 0.4885. H02 retained more variation, with corresponding values 0.4825 and 0.4036, but smaller BMI coefficients. L02's TC SD was close to the reference, 43.6778 versus 42.8466 mg/dL, although

its residual TC–HDL correlation remained weak. All three lipid laws had HDL SDs below the reference. No construction uniformly dominated across focal effects, dependence and dispersion.

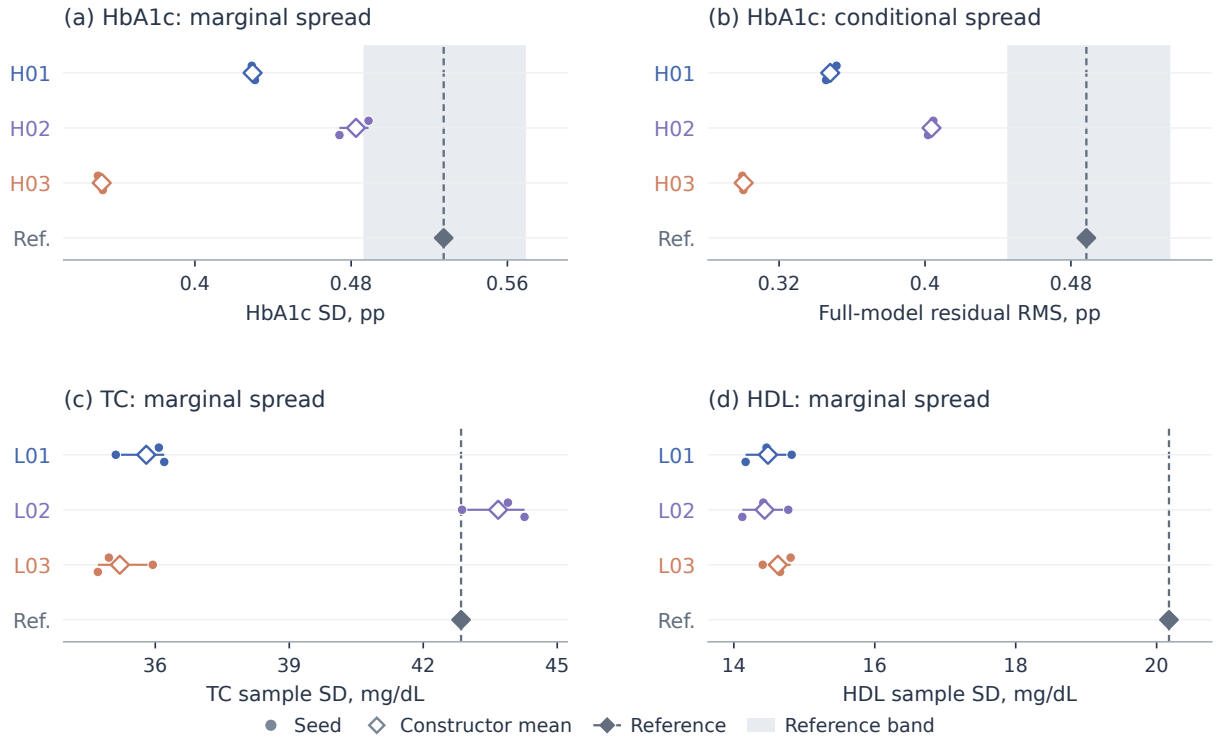


Figure S3. Dispersion across all cross-task constructions. Panels show marginal HbA1c SD, HbA1c full-model residual RMS, and marginal TC and HDL SDs for each construction and its three final draws, retaining all 36 diagnostic values from the same 18 tables. HbA1c uses the population-SD diagnostic; lipids use sample SD. These quantities measure individual or residual spread, not uncertainty in a regression coefficient. H03's relatively close adjusted BMI effect in Figure 5 coexists with reduced HbA1c spread, and L02's close TC SD coexists with weak adjusted lipid dependence. Seed ranges reflect Monte Carlo variation conditional on the selected law. Gray HbA1c bands are empirical cluster-resampling stability bands; no reference interval is supplied for lipid SDs.

The evaluation retained all 513 prespecified estimates and diagnostics, including the 54 primary outputs. Independent recomputation verified every value and all 324 matched reference comparisons; the maximum numerical difference was 5.33×10^{-14}. The remaining 189 diagnostics lacked matching cached reference estimators and remained generated-only results. Across authors, eight original studies supplied 22 author–paper uses, including context/check-only uses; optional retrieval consumed seven query credits and one original download. All 132 declared source-check entries were unavailable under the implemented semantic rules because weighting or essential operator definitions were unresolved or unsupported (Table S6). Numerical source values still informed reviewed parameter derivations, but this batch provides neither measured numerical source-fit feedback nor a before/after revision benefit. The complete results and reviews retain every endpoint, unavailable comparison and implementation incident, including correction of the independent auditor's identifier parser without changing any generated table.

Table S5. Complete Q5 cross-task primary summaries. Each row summarizes three fixed final draws from one law. All 18 author–analysis combinations are retained. HbA1c entries are percentage points (pp); lipid slopes are mg/dL per 5 kg/m². Ranges are seed minima and maxima, not confidence intervals or evidence/model uncertainty.

Request / analysis	Author	Units / increment	Reference	Generated mean	Seed min–max
HbA1c adjusted age	H01	pp / 10 years	0.075132	0.099545	0.094075–0.103606
HbA1c BMI	H01	pp / 5 kg/m²	0.083643	0.060295	0.055769–0.063938
HbA1c glucose-adjusted BMI	H01	pp / 5 kg/m²	0.056638	0.030469	0.027956–0.034291
HbA1c adjusted age	H02	pp / 10 years	0.075132	0.099539	0.098162–0.100396
HbA1c BMI	H02	pp / 5 kg/m²	0.083643	0.048923	0.029090–0.065499
HbA1c glucose-adjusted BMI	H02	pp / 5 kg/m²	0.056638	0.019363	0.002308–0.030777
HbA1c adjusted age	H03	pp / 10 years	0.075132	0.100912	0.096356–0.105015
HbA1c BMI	H03	pp / 5 kg/m²	0.083643	0.056649	0.048449–0.063758
HbA1c glucose-adjusted BMI	H03	pp / 5 kg/m²	0.056638	0.050978	0.044994–0.056901
HDL BMI	L01	mg/dL / 5 kg/m²	−4.623004	−2.186526	−2.286805–−2.118624
TC BMI	L01	mg/dL / 5 kg/m²	5.683108	10.393977	9.992186–10.676486
TC–HDL residual correlation	L01	dimensionless	0.512070	0.111234	0.108517–0.113140
HDL BMI	L02	mg/dL / 5 kg/m²	−4.623004	−2.100390	−2.194806–−2.035712
TC BMI	L02	mg/dL / 5 kg/m²	5.683108	11.260864	10.811415–11.565687
TC–HDL residual correlation	L02	dimensionless	0.512070	0.108283	0.105395–0.110194
HDL BMI	L03	mg/dL / 5 kg/m²	−4.623004	−1.986781	−2.122835–−1.913541
TC BMI	L03	mg/dL / 5 kg/m²	5.683108	10.015112	8.617345–11.698752
TC–HDL residual correlation	L03	dimensionless	0.512070	0.141343	0.121087–0.155368

HbA1c age adjusts for sex, BMI and glucose. HbA1c BMI adjusts for age/sex, with glucose added in the third analysis. Lipid BMI slopes adjust for age/sex; the residual correlation removes age/sex/BMI projections from both lipids on the same rows. H01–H03 belong only to this Q5 HbA1c request, and L01–L03 only to this Q5 lipid request; the L labels do not refer to the earlier Q2 literature arm. All 54 individual primary outputs and all 513 prespecified estimates/diagnostics remain in the complete endpoint table. The 324 matched reference

comparisons and 189 generated-only diagnostics retain their original availability labels.

Table S6. Q5 original-source uses and unavailable feedback. Eight distinct originals supplied 22 author–paper uses, including context/check-only evidence. All six authors retained their initial laws. The 132 source-check entries were unavailable, not successful numerical comparisons or failed generators. The inventory counts the three lipid originals, with the Agongo correction [28] attached to its source rather than counted as a separate original study.

Author	Original-source references	Paper uses	Unavailable source checks
H01	Pani [10]; Wisgerhof [14]; Skogberg [17]; Huang [21]; Ghazanfari [22]	5	13
H02	Pani [10]; Skogberg [17]; Huang [21]; Ghazanfari [22]	4	18
H03	Pani [10]; Skogberg [17]; Huang [21]; Ghazanfari [22]	4	32
L01	Greiner [25]; Obirikorang [26]; Agongo [27,28]	3	29
L02	Greiner [25]; Obirikorang [26]; Agongo [27,28]	3	19
L03	Greiner [25]; Obirikorang [26]; Agongo [27,28]	3	21
Total	8 distinct original studies	22	132

The evidence-use inventory and source/code review retain the source IDs, numerical roles, unresolved definitions and actual retrieval activity. Unresolved weighting and unsupported or incompletely specified operators were not replaced with different statistics. Parameter derivations can therefore use reviewed source information while the corresponding runtime source comparison remains unavailable.

S6. Dependence sensitivity and the supporting construction

The preceding open-search example acquired five seriously reviewed studies [9,10,12,13,14] without bibliographic seeds or desired signs. A fresh constructor selected a sex-stratified Gaussian copula with positive biomarker margins and generated one trial plus three complete 3,000-person tables. It retained the model after eleven approximate and 23 unavailable source comparisons per execution. Secondary BMI coefficients were 0.04217–0.05176 versus 0.05495 in the reused CHNS reference, secondary age was 0.01613–0.01771 versus 0.01530, and primary age was 0.00842–0.01084 versus 0.00811. Primary BMI/glucose were attenuated, and the female point direction and distributional tails remained imperfect. Supplementary Figure S4 retains all seven nonintercept effects and their uncertainty labels.

Before generating data or comparing targets, the source review identified an unreported BMI–glucose dependence. Two valid latent correlations, 0.10 and 0.40, implied conditional BMI coefficients of +0.03422 and −0.01095 while retaining the same margins, other bivariate laws and secondary projection. The delivered model used the declared assumption 0.25. This is an analytical sensitivity, not additional generated cohorts or a causal intervention. It demonstrates why an explicit data-generating law can identify which missing relationship matters to an intended use.

A separate analysis contextualized the same three tables with 100 community-disjoint resplits of the existing

CHNS sample, drawing 3,000 people per side and retaining all 36 metrics. Secondary BMI and age discrepancies were on scales comparable with within-sample variation, while primary glucose/female and several joint/tail discrepancies were larger. The complete table preserves the nonexchangeable real-real and generated comparisons. These are descriptive interpretation aids, not a new population or retrospectively selected equivalence cutoff.

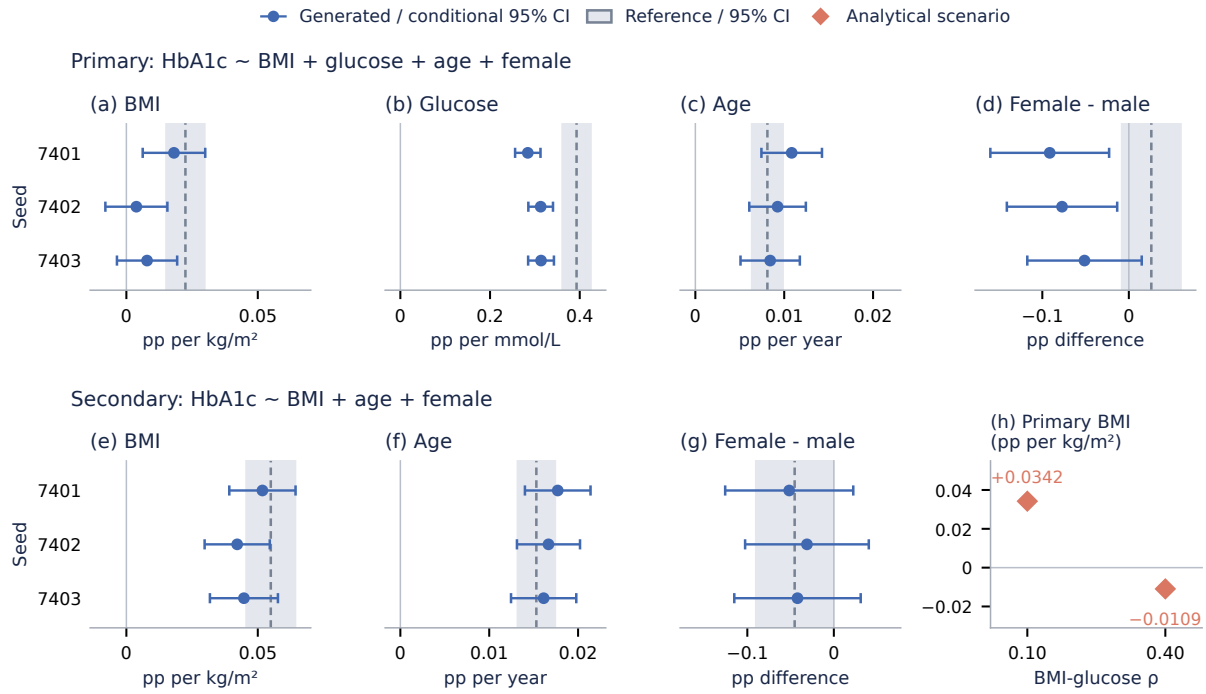


Figure S4. Analysis-specific usefulness and an unresolved joint dependence. Panels (a–d) show all primary-model nonintercept effects and (e–g) all secondary-model effects. Blue points are three fixed 3,000-person draws with conditional 95% HC3-t intervals. Gray dashed lines and bands are exposed CHNS reference estimates and 95% community-cluster CR1-t intervals; solid lines mark zero. All effects retain native units (pp: HbA1c percentage points), with shared limits for matching predictors. Generated intervals omit source, structural and transport uncertainty; overlap is not equivalence. Panel (h) displays two pre-generation analytical scenarios: latent BMI–glucose correlations 0.10 and 0.40 imply conditional BMI coefficients +0.03422 and −0.01095 while leaving marginal laws, other bivariate laws and the secondary projection unchanged. The delivered model used 0.25. These are not new draws, a fitted curve or a causal intervention. Retained initial/final stages alias the same three tables.

S7. Prospective evaluation under the earlier method

In HAALSI, eligible ages started at 40 years and the analysis used the released capillary point-of-care glucose measurement. Fasting was not an eligibility requirement. The fixed BMI and age contrasts were taken from the same additive glucose model with recorded sex. A second model estimated the difference in the BMI slope between women and men, adjusting for age. In Turkana, the two BMI slopes were adjusted for age and recorded sex. The correlation uses the residuals after adjusting both lipid measurements for age, sex and BMI.

Each case had five separately authored agent laws and one prespecified statistical generator (S01). The experiment produced 48 source-context and 36 target-context tables of 5,000 rows each. All authors retained their initial laws after source-only feedback. Initial and final labels therefore refer to the same 36 target tables. The public checkpoint preceding disclosure records work in progress. The final evaluation record documents the complete inventory, one reveal and independent recomputation of 6,704 checks. HAALSI had a

prospectively withheld target. Turkana's exact estimates were withheld, with prior investigator exposure to adjacent findings. Later Q5 reused the same 397-person Turkana reference.

Table S7. Six prespecified analyses under the earlier workflow. The columns report the measured-sample estimate, the minimum and maximum across 15 agent draws, the number agreeing in point sign, and the range of the three draws of the prescribed statistical law. The ranges describe draws under the candidate models. HAALSI glucose contrasts are in mmol/L, lipid slopes in mg/dL and the adjusted correlation is unitless.

Analysis	Measured	Agent range	Same sign	S01 range
HAALSI glucose, BMI per 5 kg/m²	0.3138	0.3285 to 0.5698	15/15	−0.0174 to 0.0180
HAALSI glucose, age per 10 years	0.2322	0.1136 to 0.3005	15/15	−0.0170 to 0.0077
HAALSI female-minus-male BMI slope per 5 kg/m²	−0.2440	−0.2487 to 0.1903	8/15	−0.0524 to 0.0046
Turkana HDL, BMI per 5 kg/m²	−4.6230	−3.5509 to −1.8173	15/15	−0.1120 to 0.0710
Turkana TC, BMI per 5 kg/m²	5.6831	10.2822 to 15.0413	15/15	−0.4657 to 0.0334
Turkana residual TC–HDL correlation	0.5121	0.1162 to 0.5361	15/15	0.3625 to 0.3840

The primary rule divided the absolute difference in each slope by the measured outcome's sample SD and used the absolute correlation difference directly. A construction passed when all nine comparisons (three analyses by three seeds) were at most 0.10. The agent counts at thresholds 0.05, 0.10, 0.20 and 0.30 were 1, 3, 5 and 5 of five for HAALSI and 0, 0, 1 and 4 for Turkana. Both S01 laws missed the primary criterion. For HAALSI, the discrepancies for three zero-valued effects were 0.0990, 0.0733 and 0.0770. The frozen decision closed the expansion. The prospective study predates the later interface for documenting numerical evidence use and supplies a separate evaluation of the underlying construction approach.

S8. Full records and retained implementation incidents

The public repository provides original code, protocols, aggregate results and independent checks. The complete v0.9 extended record preserves all six original numerical tables and full methods/results. Its Table 2 supplies every main endpoint's adjustment set, units and reference interval; Tables 3–4 retain the main and summary-assisted values; Table 5 retains all latest seven effects; Table 6 retains the real-real comparison values. These archived tables preserve the earlier analyses alongside the separately sealed iron study.

The main study, summary-assisted study, statistical supplement, HRS example, preceding open-search construction, unified workflow replication and cross-task evidence-use study preserve separate frozen identities and complete endpoint outputs. The cross-task protocol, complete endpoint table, independent numerical review and scientific interpretation document the new unit. Earlier live A/B results, including the conditioned BMI mismatch and source-replay unavailability, remain in the extended record. Figure updates and added tables consume reviewed aggregates only; manuscript integration creates no cohorts or clinical fits. In the main study, seven authors inspected exploratory outputs and kept their laws, and thirteen finalized without exploration. The two earlier live authors both kept their models after the available review.

Individual real/generated tables, third-party originals and credentials remain local; clinical reruns require lawful access and the appropriate environment. Historical model identifiers and token costs are incompletely metered. Author/affiliation/funding/conflict/ethics/access/AI declarations and final submission formatting remain to be completed. No manuscript has been submitted.

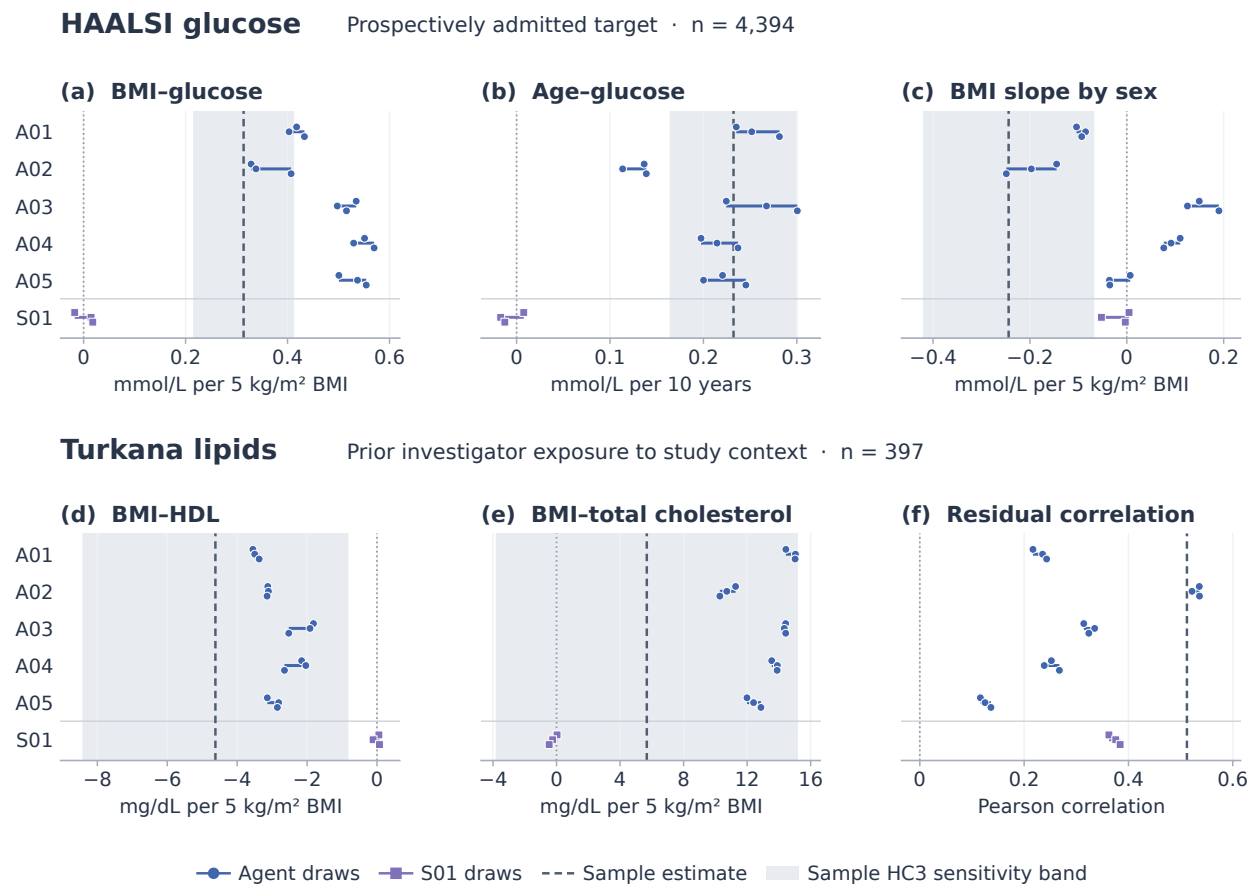


Figure S5. Earlier prospective evaluation of whole-cohort generation. Blue circles show the three draws of each agent law and purple squares the three draws of the prescribed statistical law (S01). Short segments span draws. Dashed lines mark measured-sample estimates, and shaded bands give approximate 95% HC3 sensitivity intervals for the five slopes. The residual correlation has no reference interval. HAALSI glucose main effects were directionally stable across agent draws. The sex difference in the BMI slope varied by construction. Turkana's total cholesterol point estimate had a reference interval that included zero. Its quantitative estimates were withheld before generation despite prior exposure to related published findings.

The historical supplement retains earlier enrichment and direct-row construction, the divergent-moment repair, failed transport criteria, tail defects and revision trade-offs. The v0.16 reading draft preserves the earlier layout.

S9. Iron-status analysis and independent verification

All six deliveries used two continuous columns, with three target analyses per delivery. The target survey comparison used the phlebotomy weight and 15 design degrees of freedom. Among 1,065 eligible positive-weight women, 979 had both measurements, 26 had hemoglobin only, 3 had ferritin only and 57 had neither. Thus 86 women were excluded from the paired analysis: 8.08% of eligible records and 2.47% of eligible phlebotomy-weight mass. Complete records represented 97.53% of that weight mass. These percentages describe different denominators; the weighted percentage does not remove the need for a missingness assumption. The complete-case estimates characterize the observed-pair reference under the specified weights and design. Generalizing to all eligible women requires the unavailable pairs to be represented adequately by that analysis or another justified missing-data model; the existing phlebotomy weights alone do not establish this. No additional

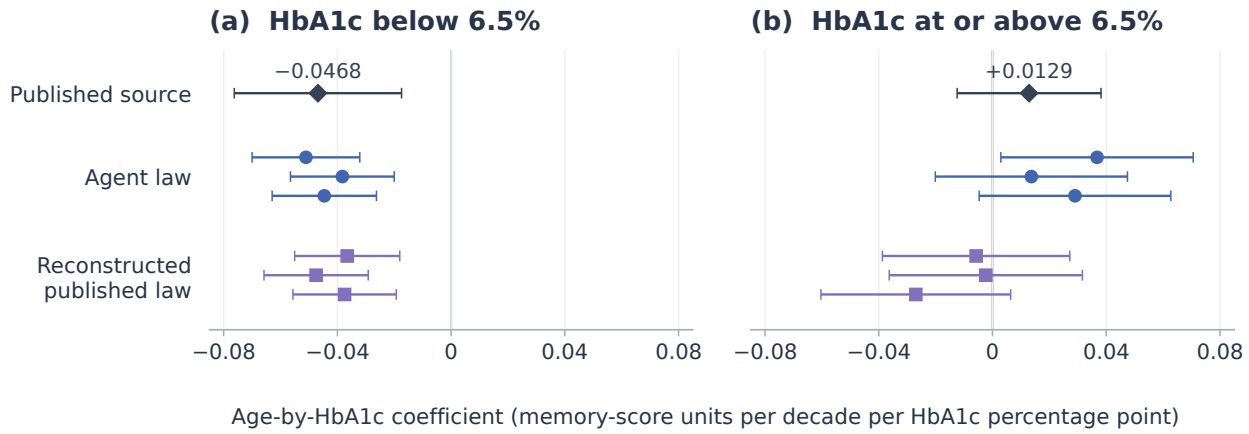


Figure S6. Two prespecified age–HbA1c interactions in a longitudinal memory model. Diamonds show the supplied coefficients; circles and squares show estimates from three draws per law (seeds 6101–6103, top to bottom). Bars show approximate 95% Wald intervals calculated from the exported source standard errors or from the generated-table fits. Units are standardized memory-score change per decade per percentage point of HbA1c within the indicated branch. Each table contains 11,509 synthetic subjects and four visits per subject. The upper-branch signs differed between constructions. The six tables describe sampling under two frozen generating laws.

weighting adjustment or imputation for missing biomarker pairs was performed. A separate implementation reproduced all 21 point estimates (three reference, eighteen generated), the six target interval endpoints and the four unweighted cell counts. The study report gives all seed values and the evaluation protocol specifies the source replay, primary log-odds-ratio comparison and secondary analyses.

S10. Reproducing a program and tracing an agent decision

In the iron task each fresh author received the same target description, source packet and clarification about a source summary. Only external source materials were allowed. The protocol specified numpy and pandas, a supplied random generator, the separate source and target contexts, and one source-only retain-or-revise opportunity. The task, evidence packet, A's final program and B's final program document the handoff. Run the programs with the recorded seeds and worker. The published code and aggregates support computational verification. Historical author-model snapshots and per-construction token counts are incompletely recorded, which limits reproduction of the original model-authoring sessions.

For the analytic part of Figure 3, let the four source counts be 173 (both low), 85 (low hemoglobin only), 331 (low ferritin only) and 1,082 (neither). Weight the first and third cells by two thirds and renormalize. Under A the final joint-low probability equals 0.8 times the reweighted joint-low probability plus 0.2 times the product of the two reweighted marginal probabilities. B keeps the original probabilities. This calculation describes the selected models and can be reproduced from published counts without sampling. The figure retains the original sealed deliveries and reference analysis alongside these model implications.

S11. A regression calculation used by one glycemia constructor

This derivation belongs to CHNS Q1, arm E1, constructor A01, selected by ordinal identifier. Jansen et al. reported two age- and sex-adjusted regressions of HbA1c [9]. One used BMI as its predictor and the other used fasting glucose. The curated packet explicitly identified them as separate regressions. The constructor wanted to draw all three variables together. A slope from a regression on BMI alone can include part of the glucose association when BMI and glucose vary together, even after their age and sex terms are removed.

Let c_B and c_G be the two reported slopes. Let s_B and s_G be the residual standard deviations of BMI and glucose after accounting for age and sex, and let r be their residual correlation. The constructor assumed a linear HbA1c response to both residual predictors, with joint slopes θ_B and θ_G and an independent error. Projecting this response onto each predictor separately gives

$$c_B = \theta_B + \theta_G r\, s_G / s_B$$

$$c_G = \theta_G + \theta_B r\, s_B / s_G$$

The reported slopes were 0.13 mmol/mol per kg/m² of BMI and 2.29 mmol/mol per mmol/L of fasting glucose. A01 used residual standard deviations of 4.07611 kg/m² and 0.49694 mmol/L and assumed a residual correlation of 0.25. Solving for the joint slopes gave 0.0642173 and 2.1583154 in the respective IFCC response units. Multiplication by 0.09148 converted these slopes to 0.00587460 and 0.19744269 in HbA1c percentage-point units [11]. This construction recovers the two source projections under its own assumptions. The target model adds a separate glucose heterogeneity component. The source-to-model trace provides the response law, the source check and the ensuing target BMI discrepancy.

S12. Implementation of the system overview

Each study uses a host coordinator to dispatch agent sessions and bind the research request to a dedicated execution path. Q4 describes the bounded open-retrieval workflow. Q5 provides a controller that binds the candidate files and a session runner that records trial feedback, decisions and closure. Its evaluator checks the closed state before reading pinned reference estimates.

In the iron study a case-specific adapter runs the agent's program with a source or target context using the isolated execution worker. The seal verifies the model and delivery hashes. The target analysis checks the sealed record before accessing the survey measurements. The execution notes document which checks were implemented in this adapter. The two studies share the separation of scientific authorship, numerical execution and evaluation, with different retrieval and feedback interfaces.

S13. Controlled check of joint-evidence construction

We examined whether explicitly requesting joint consistency added value when the necessary evidence was already available. In one artificial four-variable population, both arms received the same 15 exact summaries: marginal means and SDs, the three predictor correlations, and four coefficients from two separately adjusted regressions. These summaries determine the requested joint linear projection and residual variance, but not the full probability distribution. Two fresh constructors per arm wrote their own programs without target coefficients, a supplied solver or earlier results. The additional-instruction arm was asked to reconcile all constraints jointly and separate explained from residual variation; the ordinary arm received the same scientific task and evidence. The protocol, evaluation rules and all four programs were frozen before sampling. Each program generated three 10,000-row tables using the fixed seeds 8101–8103; no trial or feedback revision was allowed.

All four authors selected the same Gaussian law and recovered the implied joint coefficients 0.25, 0.65 and −0.40 and residual variance 0.6775. All twelve attempts succeeded. O1/C2 and O2/C1 used matching RNG layouts, so corresponding same-seed outputs were identical: twelve deliveries represented six distinct tables and one law. The three seeds within each layout gave distinct tables. In both arms, mean absolute standardized joint-coefficient errors were 0.01244, 0.01024 and 0.01337; mean absolute log residual-SD error was 0.00613. Metrics were averaged over seeds within constructors and then equally across the two constructors, without pooled-row inference.

The ordinary construction already performed the required statistical reconciliation. This bounded diagnostic therefore supplied no evidence for adding the extra instruction. It is not biomedical validation or a test of uncertain extraction, heterogeneous populations, tail behavior or feedback. The target coefficients were withheld from authors but algebraically implied by the supplied summaries; their recovery does not demonstrate learning an unconstrained relationship. The complete report retains all runs, failure checks, duplicate-output accounting and separate implementation/results audits. No additional cases or constructions were added after these results.